

Topic 5 – Communications and networks

Students should:		Curriculum reference
5.1	Understand the different types of networks (local area network (LAN), wide area network (WAN), personal area network (PAN))	CN8.1A CN9.1A
5.2	Understand why computers are connected in a network, including the Internet, and the opportunities they offer for communication and collaboration	CN9.1A, B
5.3	Understand benefits and drawbacks of the use of wired and wireless connectivity	CN7.1A, D CN8.3A
5.4	Understand that network data speeds are measured in bits per second (Mbps, Gbps)	CN7.1E
5.5	Know that data can be transmitted wirelessly using both Wi-Fi and mobile phone networks	CN7.1B
5.6	Understand that data can be transmitted in packets and that control information is provided in the packet header (sender's address, receiver's address, sequence number)	CN7.4B

Topic 5 (continued)

Students should:		Curriculum reference
5.7	Understand the risks posed to data by missing software security updates, malicious software (malware), malicious USBs/digital devices designed to intercept data	CN8.4A
5.8	Know what is meant by the term 'Internet'	CN7.3A
5.9	Know what is meant by the term 'World Wide Web'	CN7.2A
5.10	Understand the components of the WWW (web server URLs, HTTP, HTTPS, HTML)	CN8.1A
5.11	Understand the role of components used to access the Internet (modem, router, switch, wireless access point (WAP)) and how these are combined)	CN8.2A
5.12	Know the role of web browsers, search engines and filter software	CN7.3B, C, D
5.13	Know how digital devices can receive information (satellite, cable, broadcast)	CN7.1C CN8.2A
5.14	Understand why appropriate wireless communication protocols (Wi-Fi, Bluetooth®, Near-field Communication (NFC)) should be used for a particular task	CN8.3A
5.15	Know that technology enables individuals' movements and communications to be monitored (CCTV, ID cards, social media posts, GPS location data, eavesdropping)	SR7.1A CN8.4A CN8.5A